

Exam Review

HW #1 - P3b

$$e^{j\theta} = (\cos \theta + j \sin \theta)$$

$$\begin{aligned} f(t) &= 12 \cos(200t + 45^\circ) \rightarrow \vec{F} = 12 \angle 45^\circ \\ g(t) &= 5 \sin(200t + 102^\circ) \\ &= 5 \cos(200t + 102^\circ - 90^\circ) \\ &= 5 \cos(200t + 12^\circ) \rightarrow \vec{G} = 5 \angle 12^\circ \end{aligned}$$

$$h_b(t) = f(t) \cdot g(t)$$

$$\begin{aligned} &= (12 \times 5) \angle (45^\circ + 12^\circ) \\ &= 12 \cos(45^\circ) + j 12 \sin(45^\circ) = 8.4853 + j 8.4853 \\ &= 5 \cos(12^\circ) + j 5 \sin(12^\circ) = 4.8907 + j 1.0396 \end{aligned}$$

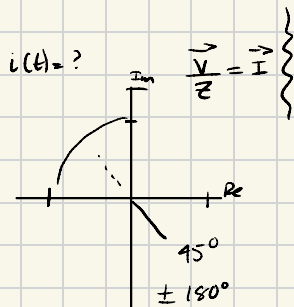
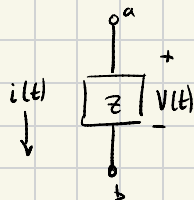
$$\Rightarrow 60 \angle 57^\circ$$

HW #1 - P5a

$$\begin{aligned} v(t) &= 12 \cos(\omega t + 20^\circ) \\ z &= -1.6971 + j 1.6971 \end{aligned}$$

$$z = \sqrt{(-1.6971)^2 + (1.6971)^2} \angle \tan^{-1}\left(\frac{1.6971}{-1.6971}\right)$$

$$2.4 \angle 135^\circ$$



$$\frac{v(t)}{z} = i(t)$$

$$\begin{aligned} \vec{I} &= \frac{\vec{V}}{z} = \frac{12 \angle 20^\circ}{2.4 \angle 135^\circ} = \frac{12}{2.4} \angle (20^\circ - 135^\circ) \\ &= 5 \angle -115^\circ \\ &\Rightarrow 5 \cos(\omega t - 115^\circ) \end{aligned}$$

HW #2 - P4

$$R = 560 \Omega$$

$$L = 100 \text{ mH}$$

$$C = 0.1 \mu\text{F}$$

$$i(0) = 0 \text{ A}$$

$$i'(0) = 1000 \text{ A/s}$$

$$I(s) = \frac{i(0)s + \left[i'(0) + \frac{R}{L} i(0) \right]}{s^2 + \underbrace{\frac{R}{L}s}_a + \underbrace{\frac{1}{LC}}_b} = \frac{m(s)}{q(s)}$$

$$a = 5600$$

$$b = 100 \cdot 10^6$$

$$s_{1,2} = \frac{-a \pm \sqrt{a^2 - 4b}}{2} = \frac{-\left(\frac{R}{L}\right) \pm \sqrt{\left(\frac{R}{L}\right)^2 - 4\left(\frac{1}{LC}\right)}}{2}$$

$$\Rightarrow = -\left(\frac{R}{2L}\right) \pm \sqrt{\left(\frac{R}{2L}\right)^2 - \frac{1}{LC}} = -2800 \pm j9600$$

$$\begin{aligned} s^2 + as + b &= (s+c)^2 + d \Rightarrow \left(s + \frac{R}{2L}\right)^2 + \left[\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right] \\ c &= \frac{a}{2} = \frac{R}{2L}, \quad d = b - \frac{a^2}{4} = \left[\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right] \end{aligned}$$

$$\Rightarrow I(s) = \frac{i(0)s + \left[i'(0) + \frac{R}{L} i(0) \right]}{s^2 + \frac{R}{L}s + \frac{1}{LC}} = A \left[\frac{s + \frac{R}{2L}}{\text{" "}} \right] + B \left[\frac{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}{\text{" "}} \right]$$

$$\Rightarrow I(s) = i(0)s + \left[i'(0) + \frac{R}{L} i(0) \right] = A \left[s + \underbrace{\frac{R}{2L}}_{2800} \right] + B \left[\underbrace{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}_{9600} \right] = A s' + \left[A 2800 + B 9600 \right] s^0$$

$$i(0) = A = 0$$

$$i'(0) + \frac{R}{L} i(0) = \cancel{A 2800} + B 9600$$

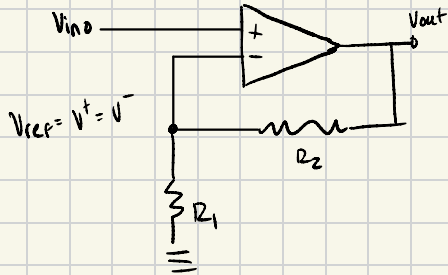
$$\Rightarrow B = \frac{i'(0)}{9600} = \frac{10,000}{9600} \approx 0.1042$$

$$I(s) = A \left[\frac{\dots}{\dots} \right] + B \left[\frac{\sqrt{\frac{1}{LC} - \left(\frac{R}{2L}\right)^2}}{\left(s + \frac{R}{2L}\right)^2 + \left[\frac{1}{LC} - \left(\frac{R}{2L}\right)^2\right]} \right] = 0.1042 \left[\frac{9600}{(s+2800)^2 + 9600^2} \right]$$

$$\mathcal{L}^{-1} \hookrightarrow i(t) = K e^{at} \sin(bt) = 0.1042 e^{-2800t} \sin(9600t) A$$

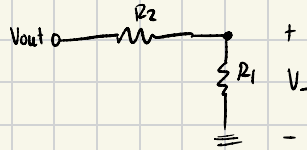
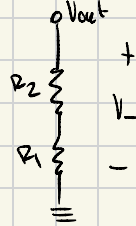
$$t \geq 0$$

Pre lab 4 - P1



$$V_- = V_+$$

$$i_- = i_+ = 0$$



$$V_+ = \frac{R_1}{R_1 + R_2} V_{out}$$

V_{in}

$$H(j\omega) = \frac{V_{out}}{V_{in}} = \frac{R_1 + R_2}{R_1}$$

$$= 1 + \frac{R_2}{R_1}$$

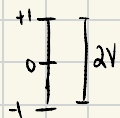
Slide 4a

8-bit ADC

$N = \# \text{ of bits}$

$$\text{levels} = 2^8 = 256 \text{ levels}$$

$$\# \text{ levels} = 2^N$$



Q = How many Volts does each level
Step size represent?

$$\frac{\text{Volts}}{\text{level}} = \frac{\text{Volt range}}{\# \text{ of levels}} = \frac{2V}{256 \text{ levels}} \approx 7.8 \text{ mV/level}$$

128	64	32	16	8	4	2	1
0	0	0	0	0	1	1	0

$$4+2 = 6 \text{ levels}$$

$$6 \text{ levels} \cdot 7.8 \text{ mV/level} = 46.8 \text{ mV}$$

$$-1V + 0.0468V = \boxed{-0.9532V}$$